

Cell Biology Basics

The Cell Theory:

1. All living organisms are composed of cells
2. The cell is the basic unit of life
3. All cells come from pre-existing cells

Cell Types:

- Prokaryotic: Simple cells without nucleus (bacteria)
- Eukaryotic: Complex cells with nucleus (plants, animals)

Cell Organelles:

- Nucleus: Contains genetic material
- Mitochondria: Powerhouse of the cell
- Ribosomes: Protein synthesis
- Endoplasmic Reticulum: Transport system